

Program 7:

Project Structure

project-7

└─ nginx-deployment.yaml

Step 1: Start Minikube

Step 2: Create Deployment File (nginx-deployment.yaml)

Step 3: Apply the Deployment

kubectl apply -f nginx-deployment.yaml

```
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~$ cd DevOps/
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps$ ls
Dockerfile  package-lock.json  program2  program4  project-6  project-8
package.json  program1            program3  program5  project-7
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps$ cd project-7
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-7$ ls
nginx-deployment.yaml
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-7$ cat nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest      # pulled from Docker Hub
          ports:
            - containerPort: 80
```

Step 4: Check Deployment and Pods

Check deployments:

kubectl get deployments

Check pods:

kubectl get pods

```
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-7$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment  0/2     2            0           13s
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-7$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment  0/2     2            0           16s
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-7$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment  0/2     2            0           17s
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-7$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment  0/2     2            0           19s
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-7$ kubectl get pods -o wide
```

Step 5: Expose the Deployment as a Service

kubectl expose deployment nginx-deployment \

--type=NodePort \

--port=80

Step 6: Check Services

kubectl get services

Step 7: Access the Web App

Use Minikube to open the service:

minikube service nginx-deployment

```
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-7$ kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP            NODE       NOMINATED NODE   READINESS GATES
nginx-deployment-6f9664446b-62zkk  1/1     Running   0          50s   10.244.0.4    minikube   <none>            <none>
nginx-deployment-6f9664446b-jfsmg  1/1     Running   0          50s   10.244.0.3    minikube   <none>            <none>
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-7$ kubectl expose deployment nginx-deployment --type=NodePort
service/nginx-deployment exposed
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-7$ kubectl get services
NAME          TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes    ClusterIP   10.96.0.1    <none>         443/TCP          12m
nginx-deployment  NodePort    10.99.67.24  <none>         80:31522/TCP     14s
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-7$ minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:31522

 Opening service default/nginx-deployment in default browser...

